Ideation Phase

Brainstorm & Idea Prioritization

Date	12 October 2026
Team ID	SWUID2026402745
Project Name	HotelPro - Hotel Room Booking and Management System
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization:

A brainstorming session gives the team room to put every idea on the table before any of them is judged. The focus is on generating many ideas, building on one another's suggestions and only then deciding which ones deserve to be built. For HotelPro, the team used this approach to turn the problems found in the SRS into a ranked list of features and a clear build order.

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team agreed on a few ground rules before starting:

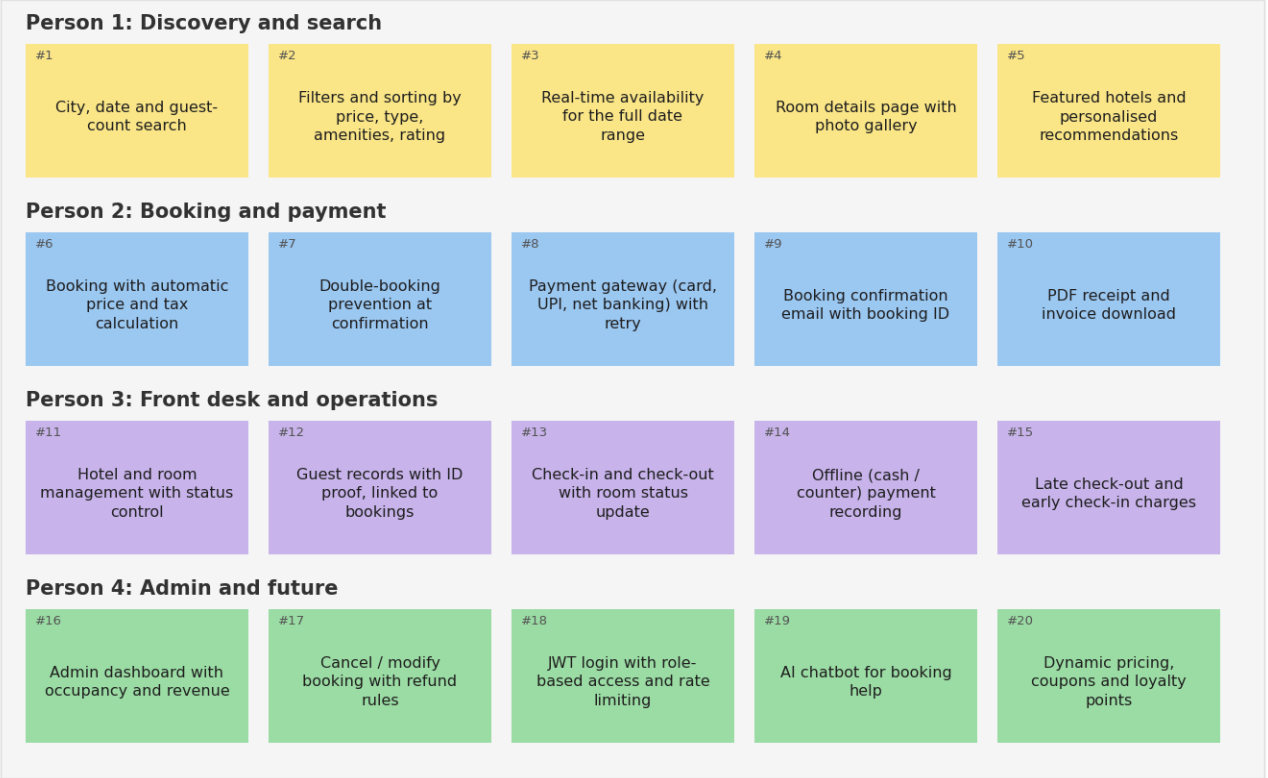
- Defer judgment: no idea is rejected while ideas are being collected.
- Go for quantity first; refine quality afterwards.
- Build on other people's ideas instead of replacing them.
- Stay tied to the problem statements and the SRS modules.

Selected problem statement (How might we...): How might we let customers find an available room, book it and pay online without double bookings or failed payments, while giving hotel staff and administrators simple tools to manage rooms, guests, check-in and check-out, refunds and reports?

Input	Source
Problems to solve	Six problem areas taken from the SRS objectives: double bookings, no real-time availability, manual guest and booking paperwork, unclear cancellations and refunds, front-desk check-in and check-out done by hand, and no admin view of occupancy and revenue
Scope reference	Software Requirements Specification: modules for registration and login, JWT authentication, hotel and room management, room search and availability, booking, booking management, guest management, check-in and check-out, payment and the admin dashboard
Team format	Four members, each taking one area: discovery and search, booking and payment, front desk and operations, admin and future ideas

Step-2: Brainstorm, Idea Listing and Grouping

Each team member silently wrote down five ideas for their area, then shared them with the group. The board below shows all 20 ideas collected.



Related ideas were then grouped into six themes:

Theme	Ideas	Count
A. Discovery and search	#1 City, date and guest-count search; #2 Filters and sorting by price, type, amenities, rating; #3 Real-time availability for the full date range; #4 Room details page with photo gallery; #5 Featured hotels and personalised recommendations	5
B. Booking and checkout	#6 Booking with automatic price and tax calculation; #7 Double-booking prevention at confirmation; #8 Payment gateway (card, UPI, net banking) with retry	3
C. Confirmation and after-booking	#9 Booking confirmation email with booking ID; #10 PDF receipt and invoice download; #17 Cancel / modify booking with refund rules	3
D. Hotel and front-desk operations	#11 Hotel and room management with status control; #12 Guest records with ID proof, linked to bookings; #13 Check-in and check-out with room status update; #14 Offline (cash / counter) payment recording; #15 Late check-out and early check-in charges	5
E. Admin, security and insights	#16 Admin dashboard with occupancy and revenue; #18 JWT login with role-based access and rate limiting	2
F. Future innovation	#19 AI chatbot for booking help; #20 Dynamic pricing, coupons and loyalty points	2

Observation: themes A and D hold the most ideas (five each), while theme B is small but contains the riskiest items (#7 and #8), because double bookings and failed payments are the core problems the SRS sets out to remove.

Step-3: Idea Prioritization

Each idea was scored from 1 to 5 for impact (how much it helps users and the business) and feasibility (how easy it is to build within the semester using React, Node.js/Express and MongoDB). The score is impact x feasibility, and the matrix places each idea in a quadrant.



Prioritized idea list (sorted by score):

ID	Idea	Impact	Feasibility	Score	Quadrant	Decision
#1	City, date and guest-count search	5	5	25	Build first	MVP - Sprint 2
#18	JWT login with role-based access and rate limiting	5	5	25	Build first	MVP - Sprint 1
#2	Filters and sorting by price, type, amenities, rating	4	5	20	Build first	MVP - Sprint 2
#3	Real-time availability for the full date range	5	4	20	Build first	MVP - Sprint 3
#6	Booking with automatic price and tax calculation	5	4	20	Build first	MVP - Sprint 3
#11	Hotel and room management with status control	5	4	20	Build first	MVP - Sprint 2
#13	Check-in and check-out with room status update	5	4	20	Build first	MVP - Sprint 5
#9	Booking confirmation email with booking ID	4	4	16	Build first	MVP - Sprint 5

ID	Idea	Impact	Feasibility	Score	Quadrant	Decision
#12	Guest records with ID proof, linked to bookings	4	4	16	Build first	MVP - Sprint 5
#4	Room details page with photo gallery	3	5	15	Quick win	Include - Sprint 2
#7	Double-booking prevention at confirmation	5	3	15	Core but complex	MVP - Sprint 3
#8	Payment gateway (card, UPI, net banking) with retry	5	3	15	Core but complex	MVP - Sprint 4
#10	PDF receipt and invoice download	3	4	12	Quick win	Include - Sprint 5
#14	Offline (cash / counter) payment recording	3	4	12	Quick win	Include - Sprint 4
#16	Admin dashboard with occupancy and revenue	4	3	12	Core but complex	MVP - Sprint 6
#17	Cancel / modify booking with refund rules	4	3	12	Core but complex	MVP - Sprint 5
#5	Featured hotels and personalised recommendations	3	3	9	Defer / future	Basic featured-hotels version - Sprint 6
#15	Late check-out and early check-in charges	2	3	6	Defer / future	Future enhancement
#20	Dynamic pricing, coupons and loyalty points	3	2	6	Defer / future	Future enhancement
#19	AI chatbot for booking help	2	1	2	Defer / future	Future enhancement

Conclusion:

- Ideas in the Build first and Core but complex quadrants form the minimum viable product: login with role-based access, hotel and room management, search and filters, real-time availability, booking with double-booking prevention, payment, check-in and check-out, cancellation with refunds, confirmation emails and the admin dashboard.
- Quick wins (room photo gallery, PDF receipts, offline payment recording) are added when sprint capacity allows.
- Recommendations start as a basic featured-hotels version; late check-out charges, dynamic pricing with coupons and loyalty points, and the AI chatbot are recorded as future enhancements, as listed in the SRS.
- Core but complex ideas, especially double-booking prevention and payment handling, receive early testing because MongoDB does not enforce relational integrity and these carry the highest risk of booking errors.